

Genre: Project

Name: _____

Class: 10

Teacher: Mr Lloyd

Issued date: W/b 27 July 2026

DUE DATE: Friday, 11 September 2026

Assessment Policy

1. The assessment is to be submitted on or before the due date.
 2. If you attend TAFE/sporting commitments/traineeships on the day the assessment is due, the assessment must be handed in to the College on or before that day.
 3. If you are ill on the day your assessment is due, you must submit your assessment on the day the assessment is due. This can be done electronically or submission of your work by another person (eg; family member) in person to the College.
 4. If there are other extenuating circumstances, which cause your absence on the day an assessment is due (death, personal or family distress), you will need to speak to your teacher and the HOTL as soon as possible before or on the due date. The policy remains that assignments are submitted on or before the due date.
 5. Ensure you submit your assessment in the approved manner – via Schoolbox, via email or hard copy. Ensure you keep a copy of your assessment throughout the drafting process and including the final submission.
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Assessment Extension Request

Extensions are only granted in exceptional circumstances. If you require an extension for this assessment item you will need to speak firstly to your teacher and secondly, the relevant Head of Department (who will consult the Head of Teaching & Learning). Any application for extension must be completed at least (3) days prior to the due date.

An extension is approved for this assessment. The new due date is: _____.

Class Teacher

HOD / HOTL

Assessment Adjustments (for Teacher use only)

The following adjustments have been made for this student in relation to this assessment item:

- Additional discussion to understand the assessment/ scaffolding/ researching/ presenting
- Extra time/ Reduced word length
- Modified assessment item
- Modified marking criteria
- Other:

Westside Christian College

Year 10 Digital Technologies



In class assignment
Semester 2, 2026

Handed out: w/b 27 July 2026
Date Due: Friday, 11 September 2026

Data Visualisation Assignment

Name: _____

Teacher: Mr Lloyd

Results

KNOWLEDGE & UNDERSTANDING	
PRODUCTION & PROCESS SKILLS	

Dataset Choice

Select data from one of the following datasets and answer the supplied questions. These are the questions that you answer using words and data visualisation on an html page. Each question requires one data visualisation at least to be used with writing answering the question posed and useful formatting.

Dataset1: Australian Road Deaths Database

<https://data.gov.au/data/dataset/australian-road-deaths-database/resource/d54f7465-74b8-4fff-8653-37e724d0ebbb>

Dataset2: Koala Data

<https://www.data.qld.gov.au/dataset/koala-hospital-data/resource/a08fb586-7ebe-45cf-97ad-5a948e26036b>

Questions to answer

Dataset 1:

Sub-questions:

- What is the most dangerous day of the week to drive?
- Is Christmas more dangerous than Easter?
- Are fatalities reducing over time?
- Should I ride a motorbike?

Overarching question – you answer with a paragraph: How safe are our roads? Make a comment about what times are not as safe as others, and whether or not your answer to the motorbike question is valid.

Dataset 2:

Sub-questions:

- What is the most common way disease or problem that Koalas presented to the hospital with?
- How many koalas hospitalised were more than 4kg - compared with all Adults?
- In which council area was the conjunctivitis cases the most? Use the following LGAs: Somerset Regional Council, Moreton Bay Regional Council, Logan City Council, Redland City Council, Brisbane City Council, Ipswich City Council, & Gold Coast City Council
- Using the suburb of Thornlands has the level of sick and injured koalas increased over the years or decreased? Use 2009, 2014, and 2019 as comparison years.

Overarching question – you answer with a paragraph: Are the numbers of koalas in the Greater Brisbane area increasing or decreasing? What does this tell you about koalas? Are there more or less sick/injured/dead koalas or what else is happening?

Required Elements

You will be constructing a small website to display the **questions** and **answers** to the questions that have been posed and that you have selected.

As part of your website (*a page of your website*) you will need to write an **evaluation**. In this page detail the *importance* of the information you have portrayed along with the *reasons* around the *advantages* of adding charts to the information you have displayed. Refer to the idea that charts add to enhancement of the information.

Stage 1

The first stage is to investigate the data and develop four charts that answer four of the questions. Ensure that you also answer the overarching question.

Stage 2

Develop each webpage that you need and add in the content and the charts (following your designs). Ensure you also reference where the data comes from in your webpages – that is, provide a link to the site and information regarding the data. Ensure you answer the overarching question – using the data you have shown and what you think the data shows overall.

Stage 3

Write a separate webpage for the evaluation *considering* the requirements of the evaluation.

Method of submission: You will either need to transfer all files to your webserver in a folder called 'data'

OR

compress the complete folder and submit to SchoolBox. This will ensure that all elements of your websites are available on the web, including any images and datasets.

Data Visualisations Rubric

	ChartJS scripting
Above Satisfactory	Students effectively use ChartJS data libraries to provide visualisations on a variety of data which may explain correlations and probable causation. They interrogate data in innovative or interesting ways.
Satisfactory	Students provide a simple graph and table demonstrating some use of the ChartJS CDN.
Below Satisfactory	Students supply coding which does not work. Data visualisations do not contain items such as explanations or labels.

By the end of Year 10 students develop and modify innovative digital solutions, decompose real-world problems, and critically evaluate alternative solutions against stakeholder elicited user stories. Students acquire, interpret and model complex data with databases and represent documents as content, structure and presentation. They design and validate algorithms and implement them, including in an object-oriented programming language. Students explain how digital systems manage, control and secure access to data; and model cyber security threats and explore a vulnerability. They use advanced features of digital tools to create interactive content, and to plan, collaborate on, and manage agile projects. Students apply privacy principles to manage digital footprints.

ACHIEVEMENT STANDARD ASPECT with Relevant Content Descriptions

Students acquire, interpret and model complex data with databases and represent documents as content, structure and presentation.

- represent documents online as content (text), structure (markup) and presentation (styling) and explain why such representations are important – AC9TDI10K02
- analyse and visualise data interactively using a range of software, including spreadsheets and databases, to draw conclusions and make predictions by identifying trends and outliers – AC9TDI10P02

Students develop and modify innovative digital solutions, decompose real-world problems, and critically evaluate alternative solutions against stakeholder elicited user stories.

- design and prototype the user experience of a digital system – AC9TDI10P07
- evaluate existing and student solutions against the design criteria, user stories, possible future impact and opportunities for enterprise – AC9TDIP10

Students design and validate algorithms and implement them, including in an object-oriented programming language.

- implement, modify and debug modular programs, applying selected algorithms and data structures, including in an object-oriented programming language – AC9TDI10K09

Mapping between Content Descriptions and QCAA Standard Elaborations

Knowledge & Understanding

AC9TDI10K02: represent documents online as content (text), structure (markup) and presentation (styling) and explain why such representations are important

Evidenced by: website design using HTML, CSS and content.

Processes and Production Skills

AC9TDI10P02: analyse and visualise data interactively using a range of software, including spreadsheets and databases, to draw conclusions and make predictions by identifying trends and outliers.

Evidenced by: building charts that show data in an interactive and informative way to assist in conclusions drawn from the data.

AC9TDI10P07: design and prototype the user experience of a digital system.

Evidenced by: well set out webpages, organising the charts and information in appropriate ways for ease of reading and understanding.

AC9TDI10P09: implement, modify and debug modular programs, applying selected algorithms and data structures, including in an object-oriented programming language.

Evidenced by: use and application of supplied modular JavaScript programs that provide charts to html pages.

AC9TDI10P10: evaluate existing and student solutions against the design criteria, user stories, possible future impact and opportunities for enterprise.

Evidenced by: evaluative comments about the importance of the information portrayed along with the reasons around any advantages of adding charts to information (creating knowledge).

		A	B	C	D	E
The folio of student work contains evidence of the following:						
Knowledge and understanding	Digital systems	<u>considered</u> explanation of how digital systems manage, control and secure access to data	<u>detailed</u> explanation of how digital systems manage, control and secure access to data	explanation of how digital systems manage, control and secure access to data	<u>description</u> of how digital systems manage, control <u>and/or</u> secure access to data	<u>statement/s about</u> data management, control and/or secure access to data
	Data representation AC9TDI10K02, 3	<u>reasoned</u> representation of documents as content, structure and presentation	<u>effective</u> representation of documents as content, structure and presentation	representation of documents as content, structure and presentation	<u>partial</u> representation of documents as content, structure <u>and/or</u> presentation	<u>directed</u> representation of documents as content, structure and/or presentation
Processes and production skills	Acquiring, managing data and analysing data AC9TDI10P01-03	<u>proficient</u> acquisition, interpretation and modelling of complex data with databases	<u>effective</u> acquisition, interpretation and modelling of complex data with databases	acquisition, interpretation and modelling of complex data with databases	<u>partial</u> acquisition, interpretation <u>and/or</u> modelling of complex data with databases	<u>fragmented</u> acquisition, interpretation and/or modelling of data with databases
	Investigating and defining AC9TDI10P04	<u>reasoned</u> decomposition of real-world problems	<u>effective</u> decomposition of real-world problems	decomposition of real-world problems	<u>partial</u> decomposition of real-world problems	<u>statement/s about</u> real-world problems
	Generating and designing AC9TDI10P05-08	<u>considered</u> design and validation of algorithms	<u>effective</u> design and validation of algorithms	design and validation of algorithms	<u>guided</u> design and validation of algorithms	<u>directed</u> design <u>and/or</u> validation of algorithms

		A	B	C	D	E
		considered development and modification of innovative digital solutions	effective development and modification of innovative digital solutions	development and modification of innovative digital solutions	partial development and modification of innovative digital solutions	fragmented development and/or modification of digital solutions
	Producing and implementing AC9TDI10P09	proficient implementation of algorithms, including in an object-oriented programming language	effective implementation of algorithms, including in an object-oriented programming language	implementation of algorithms, including in an object-oriented programming language	partial implementation of algorithms, including in an object-oriented programming language	directed implementation of algorithms
	Evaluating AC9TDI10P10	discerning critical evaluation of alternative solutions against stakeholder elicited user stories	effective critical evaluation of alternative solutions against stakeholder elicited user stories	critical evaluation of alternative solutions against stakeholder elicited user stories	partial evaluation of alternative solutions against stakeholder elicited user stories	identification of alternative solutions against stakeholder elicited user stories
	Collaborating and managing AC9TDI10P11-12	proficient use of advanced features of digital tools to: - create interactive content - plan, collaborate on and manage agile projects	effective use of advanced features of digital tools to: - create interactive content - plan, collaborate on and manage agile projects	use of advanced features of digital tools to: - create interactive content - plan, collaborate on and manage agile projects	variable use of advanced features of digital tools to partially: - create interactive content - plan, collaborate on and/or manage agile projects	directed use of digital tools
	Privacy and security AC9TDI10P13-14	discerning modelling of cyber security threats and exploration of a vulnerability	plausible modelling of cyber security threats and exploration of a vulnerability	modelling of cyber security threats and exploration of a vulnerability	partial modelling of cyber security threats and/or exploration of a vulnerability	statement/s about cyber security threats
		proficient application of privacy principles to manage digital footprints.	informed application of privacy principles to manage digital footprints.	application of privacy principles to manage digital footprints.	partial application of privacy principles to manage digital footprints.	directed application of privacy principles to manage digital footprints.